Charge densities

- At the atomic scale, the charge distribution is discrete
- In electromagnetics, we are interested in macroscopic scale where the discontinuous nature of charge distribution can be ignored
- Consider an elemental volume Δv with charge Δq centered at point P(x,y,z), the **volume charge density** at P(x,y,z) is

$$\rho_v(x,y,z) = \lim_{\Delta v \to 0} \frac{\Delta q}{\Delta v} = \frac{dq}{dv} \qquad \left(C/m^3\right)$$

- In general, ρ_v is a function of the space coordinate (x, y, z)
- It represents the average charge per unit volume for a volume Δv centered at point P
- The variation of ρ_v with spatial location is called spatial charge distribution

Charge densities (cont'd.)

- The total charge contained in a volume v is given by

$$Q = \int_v \rho_v dv$$

- In some cases, electrical charges may be distributed across the surface of a material, in which the relevant quantity of interest is the **surface charge density** ρ_s

$$\rho_s = \lim_{\Delta s \to 0} \frac{\Delta q}{\Delta s} = \frac{dq}{ds} \qquad \left(C/m^2\right)$$

  - Δq is the charge present across an elemental surface area Δs

- Similarly, if the charge is distributed along a line, we characterize the distribution in terms of the **line charge density** ρ_l

$$\rho_l = \lim_{\Delta l \to 0} \frac{\Delta q}{\Delta l} = \frac{dq}{dt} \qquad \left(C/m\right)$$

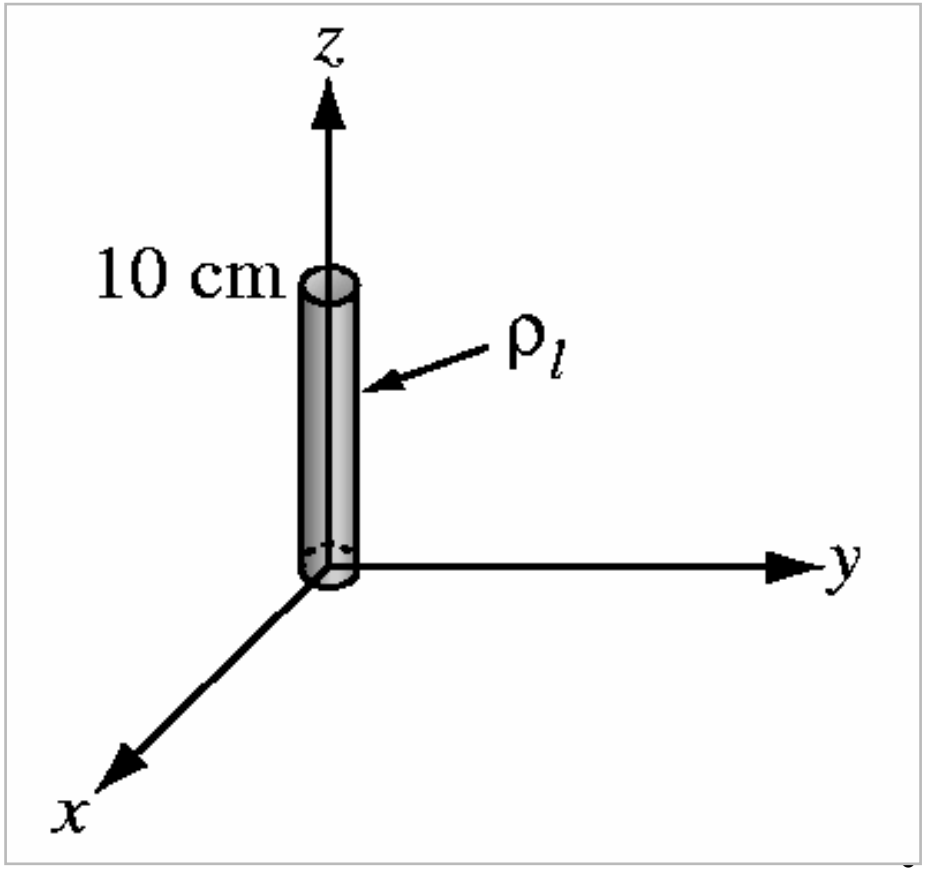
Example

Calculate the total charge Q contained in a cylindrical tube of charge oriented along the z-axis as shown in the figure. The line charge density is $\rho_l=2z$, where z is the distance in meters from the bottom end of the tube. The tube length is 10 cm.

Solution:

The total charge Q is

$$Q = \int_0^{0.1} \rho_l dz = \int_0^{0.1} 2z dz$$
$$= z^2 \Big|_0^{0.1} = 0.01\ C$$



Coulomb's law

- Coulomb's law states that the force F between two point charges Q_1 and Q_2 in free space is:
  - Along the line joining them
  - Directly proportional to the product Q_1Q_2 of the charges
  - Inversely proportional to the square of the distance R between them

$$F = \frac{kQ_1Q_2}{R^2} \quad \text{N}$$

where

$$k = \frac{1}{4\pi\varepsilon_0} \cong 9\times10^9 \, m/F$$

$$\varepsilon_0 = 8.854\times10^{-12} \cong \frac{10^{-9}}{36\pi} F/m$$

Coulomb's law (con'td.)

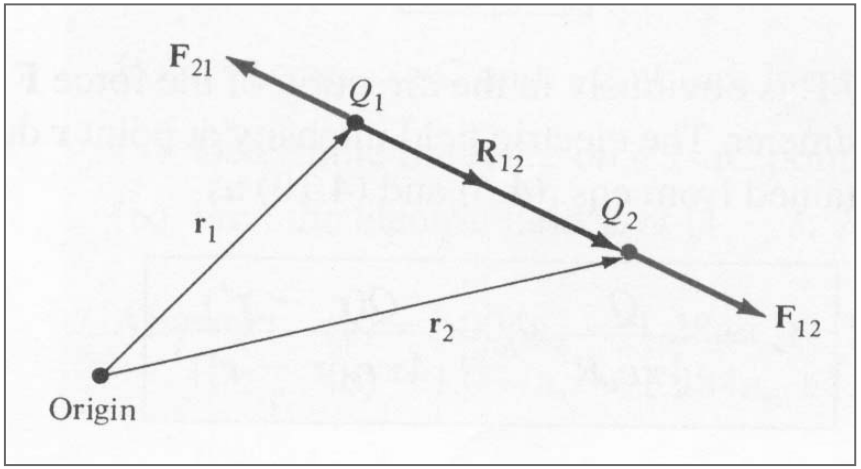
- If point charges Q_1 and Q_2 are located at points having position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$, then the force $\mathbf{F}_{12}$ on Q_2 due to Q_1 is

$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}\mathbf{a}_{R_{12}}$$

where

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1$$

$$\mathbf{a}_{R_{12}} = \frac{\mathbf{R}_{12}}{|\mathbf{R}_{12}|}$$



- Therefore we may write $\mathbf{F}_{12}$ as

$$\mathbf{F}_{12} = \frac{Q_1 Q_2 (\mathbf{r}_2 - \mathbf{r}_1)}{4\pi\varepsilon_0 |\mathbf{r}_2 - \mathbf{r}_1|^3}$$

  - $\mathbf{F}_{21}$ has the same magnitude and opposite direction as $\mathbf{F}_{12}$

Coulomb's law (con'td.)

- If Q_1 is located at the origin, then

$$\mathbf{F}_{12} = \frac{Q_1 Q_2 \mathbf{R}}{4\pi\varepsilon_0 |\mathbf{R}|^3}$$

Where **R** is a vector pointing from origin to Q_2

- **Electric field intensity** (electric field strength) **E** is the force per unit charge when placed in the electric field

$$\mathbf{E}_1 = \frac{\mathbf{F}_{12}}{Q_2} = \frac{Q_1(\mathbf{r}_2 - \mathbf{r}_1)}{4\pi\varepsilon_0 |\mathbf{r}_2 - \mathbf{r}_1|^3} \text{ (V/m)}$$

or generally speaking

$$\mathbf{E} = \frac{\mathbf{F}}{Q} \text{ (V/m)}$$

Example

EXAMPLE 3–1 Determine the electric field intensity at $P(-0.2, 0, -2.3)$ due to a point charge of $+5$ (nC) at $Q(0.2, 0.1, -2.5)$ in air. All dimensions are in meters.

Solution The position vector for the field point P

$$\mathbf{R} = \overrightarrow{OP} = -\mathbf{a}_x 0.2 - \mathbf{a}_z 2.3.$$

The position vector for the point charge Q is

$$\mathbf{R}' = \overrightarrow{OQ} = \mathbf{a}_x 0.2 + \mathbf{a}_y 0.1 - \mathbf{a}_z 2.5.$$

The difference is

$$\mathbf{R} - \mathbf{R}' = -\mathbf{a}_x 0.4 - \mathbf{a}_y 0.1 + \mathbf{a}_z 0.2,$$

which has a magnitude

$$|\mathbf{R} - \mathbf{R}'| = [(-0.4)^2 + (-0.1)^2 + (0.2)^2]^{1/2} = 0.458 \text{ (m)}.$$

we obtain

$$\mathbf{E}_P = \left(\frac{1}{4\pi\epsilon_0}\right)\frac{q(\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3}$$

$$= (9 \times 10^9)\frac{5 \times 10^{-9}}{0.458^3}(-\mathbf{a}_x 0.4 - \mathbf{a}_y 0.1 + \mathbf{a}_z 0.2)$$

$$= 214.5(-\mathbf{a}_x 0.873 - \mathbf{a}_y 0.218 + \mathbf{a}_z 0.437) \quad \text{(V/m)}.$$

The quantity within the parentheses is the unit vector $\mathbf{a}_{QP} = (\mathbf{R} - \mathbf{R}')/|\mathbf{R} - \mathbf{R}'|$, and $\mathbf{E}_P$ has a magnitude of 214.5 (V/m).

Gauss's law and its application

- Gauss's law asserts that the total **outward** flux of the **E** field over any **closed surface** in **free space** is equal to the total charge enclosed in the surface divided by ε_0

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\varepsilon_0}$$

- The total charge enclosed in the surface

$$Q = \int_V \rho_v dv$$

- Using the divergence theorem

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \int_V \nabla \cdot \mathbf{E} dv = \int_V \frac{\rho_v}{\varepsilon_0} dv$$

Electric flux density

- Therefore,

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\varepsilon_0}$$

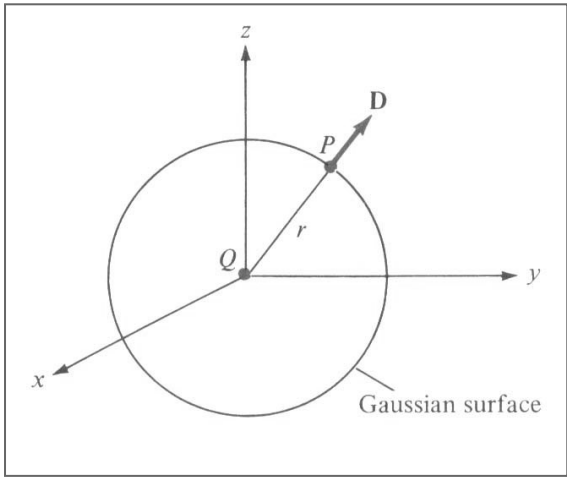
- Define a vector $\mathbf{D} = \varepsilon_0 \mathbf{E}$

$$\nabla \cdot \mathbf{D} = \rho_v \qquad (1)$$

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$$

- Equation (1) is the **first** of **Maxwell's equations for static EM fields**
- **D** is called the **electric flux density**, unit C/m^2
  - Historically, it is also called electric displacement

Example



Electric field of a point charge

- Suppose the point charge is at origin
- Choose a spherical surface

containing Q (**Gaussian surface**),

spherical coordinates are chosen

- **E** is everywhere normal to the

Gaussian surface. Moreover **E** is a function of the distance from Q only, thus **E** is a constant over the Gaussian surface

$\mathbf{E} = \mathbf{a}_r E_r$

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \oint_S E_r ds = E_r \oint_S ds = E_r 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$\mathbf{E} = \mathbf{a}_r E_r = \mathbf{a}_r \frac{Q}{4\pi\varepsilon_0 r^2}$$

Example

Determine the electric field intensity of an infinite planar charge with a uniform surface charge density ρ_s

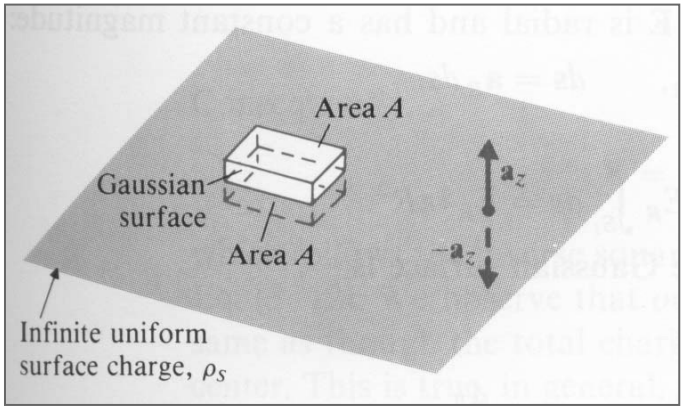
Solution:

- Due to symmetry, the **E** field is normal to the sheet
- Gaussian surface is chosen as a rectangular box with top and bottom faces of arbitrary area A **equidistant** from the planar charge
  - The **E** field on the top and bottom faces are equal in magnitude and opposite in direction
- The sides of the boxes are perpendicular to the plane
  - Direction of the sides is normal to the **E** field
  - Surface integral of **E** on the sides is zero
- On the top face

$\mathbf{E}\cdot d\mathbf{s} = (\mathbf{a}_z E_z)\cdot(\mathbf{a}_z ds) = E_z ds$

- On the bottom face

$\mathbf{E}\cdot d\mathbf{s} = (-\mathbf{a}_z E_z)\cdot(-\mathbf{a}_z ds) = E_z ds$



Example (cont'd)

– Therefore,

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = 2E_z \int_A ds = 2E_z A = \frac{\rho_s A}{\varepsilon_0}$$

$$E_z = \frac{\rho_s}{2\varepsilon_0}$$

– It can be obtained

$$\mathbf{E} = \mathbf{a}_z E_z = \mathbf{a}_z \frac{\rho_s}{2\varepsilon_0} \qquad z > 0$$

$$\mathbf{E} = -\mathbf{a}_z E_z = -\mathbf{a}_z \frac{\rho_s}{2\varepsilon_0} \qquad z < 0$$